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1. Introduction

The design and simulation, in this project, involves a high gain, low power open-loop voltage amplifier using MOSFET technology. The main goal was to obtain a minimum of 60 dB of voltage gain but definitely within a 1 mW DC power and a 500 kHz bandwidth constraint. The report is verified with KiCad to compute the trade-offs between the sizing of the transistor, biasing stability, and output swing.

2. Design Architecture

In contrast to multi-stage common-source design, this design employs Telescopic Cascode architecture and then a Source Follower buffer.

Stage of Gain: Telescopic Cascode.

A cascode topology was selected to achieve a 60 dB goal in one high-impedance stage. The gain transistors (NMOS) of Q1 and Q2 are stacked above the PMOS active loads (Q4, Q5) so that the exponentially increased output resistance is obtained. This enables the design to achieve the gain target without power penalty of several high-current stages.

Stage: Buffer Follower.

To isolate the high-impedance gain node and the 10k load a Common-Drain (Q3) buffer is used. This would allow the load to not attenuate the voltage gain significantly and give the necessary current to power the required 12pF capacitive load.

DC Biasing

The gate biases are achieved by two high-impedance voltage divider networks. The resistors of the mega-ohm range were selected to enable the biasing networks themselves

to take minimal power (around 14uA) and leave most of the 1 mW signal path budget to the active signal path

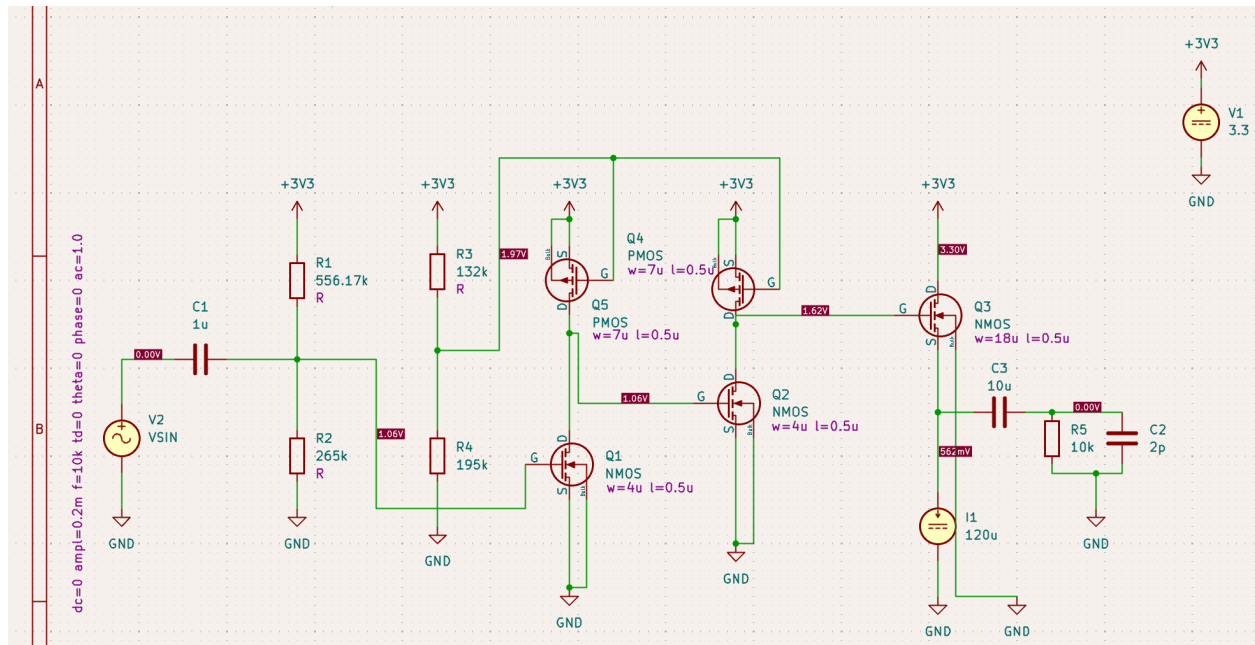


Figure 1: 3-stage [CS-CS-CD] MOSFET Amplifier Circuit Diagram (loaded)

3. Theoretical Calculations

Power & Current Constraints

Total Power $\rightarrow P_{DC} \leq 1 \text{ mW}$

Total Current Limit $\rightarrow I_{tot} = \frac{P_{DC}}{V_{DD}} = \frac{1 \text{ mW}}{3.3 \text{ V}} = 303 \mu\text{A}$

Design Allocation / Choice $\rightarrow$ Gain Stage Branch (I_{D2}) = $77 \mu\text{A}$
 $\rightarrow$ Buffer Branch (I_{D3}) = $120 \mu\text{A}$
 $\rightarrow$ Biasing Resistor Networks $\approx 14 \mu\text{A}$
 $\rightarrow$ Sum = $211 \mu\text{A} \leq 303 \mu\text{A}$ ✓

Transistor Sizing (W) & Gate Voltages (V_G)

$$I_D = \frac{1}{2} K'_n \frac{W}{L} (V_{GS} - V_{th})^2, L = 0.5 \mu\text{m}, K'_n = 200 \mu\text{A/V}^2, V_{th} = 0.5 \text{V}$$

NMOS Gain Stage (Q_1, Q_2): Target $V_{ov} = 0.31 \text{V}$

$$V_{GS1} = V_{th} + V_{ov} = 0.5 \text{V} + 0.31 \text{V} = 0.81 \text{V}$$

$$77 \mu\text{A} = \frac{1}{2} (200 \mu) \left(\frac{W_1}{0.5 \mu} \right) (0.31)^2$$

$$77 = 19.22 (W_1) \rightarrow W_1 = 4 \mu\text{m} = W_2$$

* We need to define a safety margin (V_{margin}) $\approx 0.2 \text{V}$ to 0.3V , in order to maximize output swing range & ensure stability.

$$V_{GS1, final} = V_{GS1, theo} + V_{margin}$$

$$V_{GS1, final} = 0.81 \text{V} + 0.25 \text{V} = 1.06 \text{V}$$

PMOS Active Load (Q_4, Q_5): Target $V_{ov} = 0.47 \text{V}$

$$K'_p = 100 \mu\text{A/V}^2, I_D = 77 \mu\text{A}, V_{th} = 0.5 \text{V}$$

$$77 \mu\text{A} = \frac{1}{2} (100 \mu) \left(\frac{W_4}{0.5 \mu} \right) (0.47)^2$$

$$77 = 11.045 (W_4) \rightarrow W_4 = 7 \mu\text{m} = W_5 \rightarrow \text{Ratio of } W = 7 \mu\text{m} \text{ \& } L = 0.5 \mu\text{m} = 14$$

$$I_D = \frac{1}{2} K'_p \left(\frac{W}{L} \right) (V_{SG} - V_{th})^2$$

$$77 = \frac{1}{2} (100 \mu) (14) (V_{SG} - 0.5)^2$$

$$77 = 700 (V_{SG} - 0.5)^2$$

$$(V_{SG} - 0.5)^2 = 0.11$$

$$V_{SG} - 0.5 = \sqrt{0.11}$$

$$V_{SG} - 0.5 = 0.33 \text{V}$$

$$V_{SG} \approx 0.83 \text{V}$$

∞ PMOS is connected to V_{DD} (3.3V),

$$V_{SG} = V_S - V_G$$

$$0.83 = 3.3 - V_{G4}$$

$$V_{G4} = 2.47 \text{V}$$

While 2.47 is the theoretical value, 1.97 was used in the final design because small differences between NMOS & PMOS channel lengths cause the output node to drift toward the rails.

The main goal was to "center" the output node at approx 1.62V (midpoint for 1.5V_{pp} swing), so the PMOS gate was biased slightly lower (1.97V) to increase its strength relative to the NMOS pull.

$$\infty, V_{G4} = 1.97$$

Buffer Stage (Q3)

$R_{out} \approx 1/g_m$ needs to be low enough that the $10k\Omega$ load results in $< 10\%$ drop

Required g_m : $0.90 \leq \frac{10k}{10k + (1/g_m)} \rightarrow g_m \geq 0.9mA/V$

$$g_m = \sqrt{2K'_n (W/L) I_D} \rightarrow 0.9mA/V = \sqrt{2(200\mu)(W/0.5\mu)(120\mu)}$$

$$(0.9mA/V)^2 = 0.000096 (W_3) \rightarrow W_3 \approx 8\mu m$$

In order to account for the body effect ($n \approx 0.2$) which reduces effective transconductance
so, $W_3 = 18\mu m$

Resistor Value Derivations

NMOS Divider (R_1, R_2)

$$V_{G1} = 1.06V \quad R_{in} \geq 100k\Omega$$

$$\text{Set } R_{tot} = 821.17k\Omega$$

$$R_2: V_{G1} = 3.3 \frac{R_2}{R_{tot}} \rightarrow R_2 = 265k\Omega$$

$$R_1: R_1 = 821.17k\Omega - 265k\Omega = 556.17k\Omega$$

$$R_{in}: R_1 \parallel R_2 = \frac{556k\Omega \cdot 265k\Omega}{821k\Omega} = 179.5k\Omega \geq 100k\Omega$$

PMOS Divider (R_3, R_4)

$$V_{G4} = 1.97V$$

$$\text{Set } R_{tot} = 327k\Omega$$

$$R_4: 1.97 = 3.3 \frac{R_4}{327k\Omega} \rightarrow R_4 = 195k\Omega$$

$$R_3: R_3 = 327k\Omega - 195k\Omega = 132k\Omega$$

Frequency Response (Bandwidth)

Need to show that the buffer drives the total load capacitance ($C_L = 10pF + 2pF = 12pF$) past 500 kHz

$$R_{out} \approx \frac{1}{g_{m3}} = \frac{1}{1.3mA/V} \approx 769\Omega$$

$$f_H = \frac{1}{2\pi R_{out} C_L} = \frac{1}{2\pi (769)(12pF)} = 17.2MHz \geq 500kHz$$

Gain (Q1 and Q2):

$$g_{m1} = g_{m2} = \frac{2I_D}{V_{ov}} = \frac{2(77)}{0.31} = 496.77\mu S$$

$$r_{on} = r_{op} = \frac{1}{\lambda I_D} = \frac{1}{(0.04)(77)} = 324.68k\Omega$$

$$R_{out1} = R_{out2} = r_{on} || r_{op} = 162.34k\Omega$$

$$A_{v1} = A_{v2} = -g_m \cdot R_{out} = -(496.77)(162.34) = -80 V/V$$

$$A_{dB} = 20\log_{10}(80) = 38dB \leq 40dB$$

(Q4 and Q5):

Verify Load Sensitivity (Gain reduction $\leq 10\%$):

$$R_L = 10k\Omega$$

$$r_{o3} = \frac{1}{(0.04)(120\mu A)} = 208.3k\Omega$$

$$A_{v3} \text{ (loaded gain)} = \frac{R_L || r_{o3}}{(R_L || r_{o3}) + \frac{1}{g_{m3}}} = \frac{9.54k\Omega}{9.54k\Omega + 1k\Omega} = 0.905 V/V$$

$$\text{Gain Reduction: } (1 - 0.905) \cdot 100 = 9.5\% \leq 10\%$$

Total Gain:

$$A_{v, total} = A_{v1} \cdot A_{v2} \cdot A_{v3} = (-80)(-80)(1) = 6400 V/V$$

$$A_{dB, total} = 20\log(6400) = 76dB \geq 60dB$$

4. Experimental Results / Simulation

Table 1: Results for Theoretical vs. KiCad Simulation

	Parameters	Constraints	Theoretical Results	Experimental Results	Pass? (Y/N)
Input	Supply Voltage	3.3V	3.3V	3.3V	Y
	Signal Amplitude	$\leq 10 \text{ mVpp}$	—	0.2 mVpp	Y
	Input Resistance	$\geq 100k\Omega$	179.5 k Ω	179.5 k Ω	Y
Gain	Total open-loop Gain (unloaded)	$\geq 60dB$ ($\geq 1000V/V$)	76dB	71.1dB	Y

	Total open-loop Gain (loaded)	—	—	70.4dB	—
	Gain Reduction	$\leq 10\%$	9.5%	6.69%	Y
	Gain per stage	$\leq 40\text{dB}$	Stages 1 & 2: 38dB Stage 3: 0dB	Stage 1: 36dB Stage 2: 31dB Stage 3: -1.7dB	Y
Bandwidth	-3 dB Bandwidth	$\geq 500\text{kHz}$	17.2 MHz	690 kHz	Y
Output	Output Swing	$\geq 1.5\text{Vpp}$	1.6Vpp	1.52Vpp	Y
Power	Total DC Power	$\leq 1\text{mW}$	0.97mW	0.949mW	Y
	Max branch current	$\leq 200\mu\text{A}$	120 μA	120 μA	Y
	Transistor Length	$0.5\mu\text{m} \leq L \leq 5\mu\text{m}$	0.5 μm	0.5 μm	Y
	Overdrive Voltage	$0.2\text{V} \leq V_{ov} \leq 0.6\text{V}$	$V_{ovn} = 0.31\text{V}$ $V_{ovp} = 0.47\text{V}$	$V_{ovn} = 0.33\text{V}$ $V_{ovp} = 0.49\text{V}$	Y

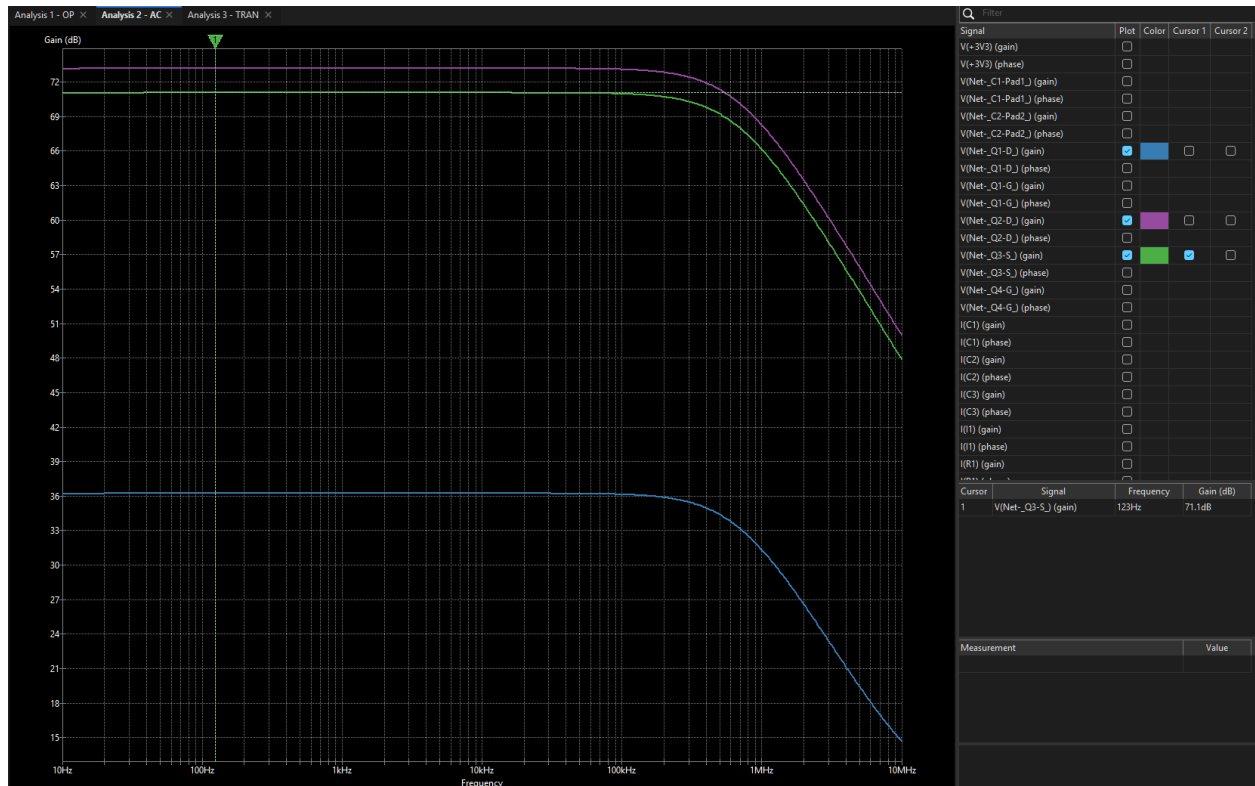


Figure 2: AC analysis for unloaded gain ($R_{out} = \infty$)

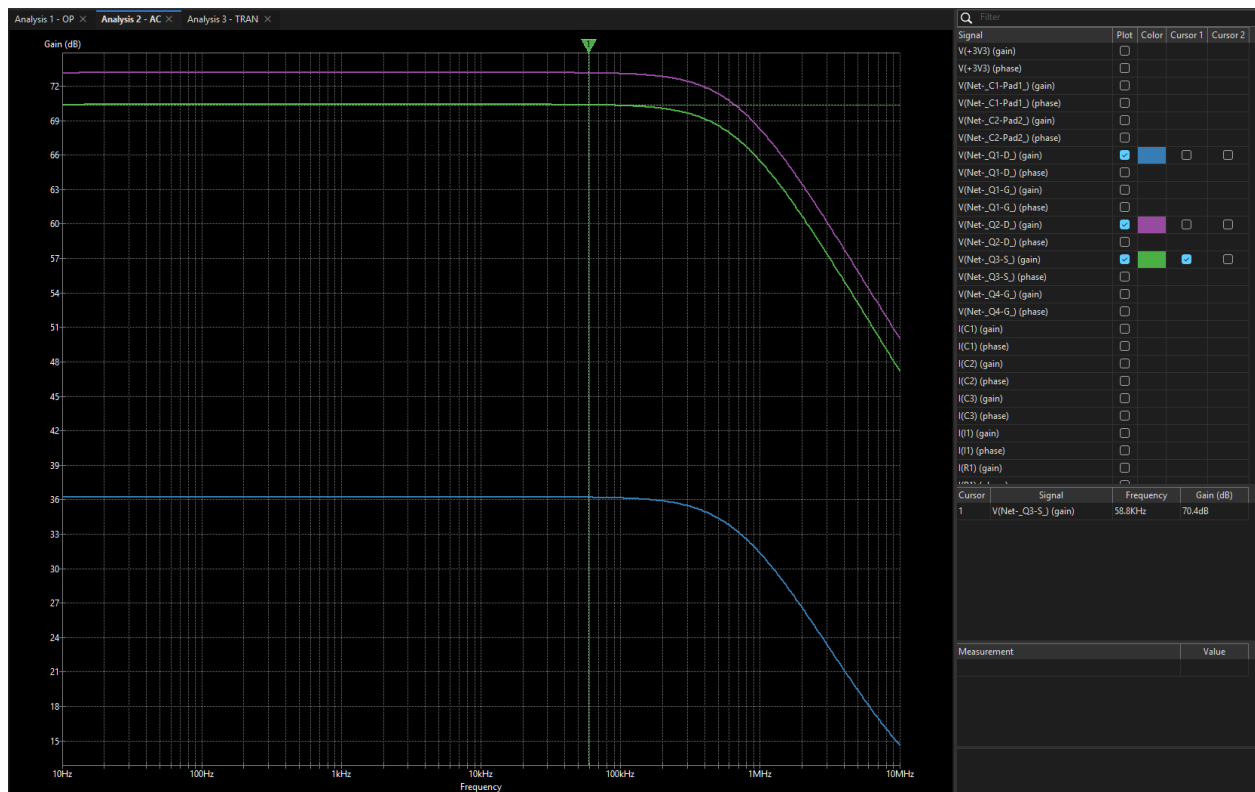


Figure 3: AC analysis for loaded gain ($R = 10k\Omega$)

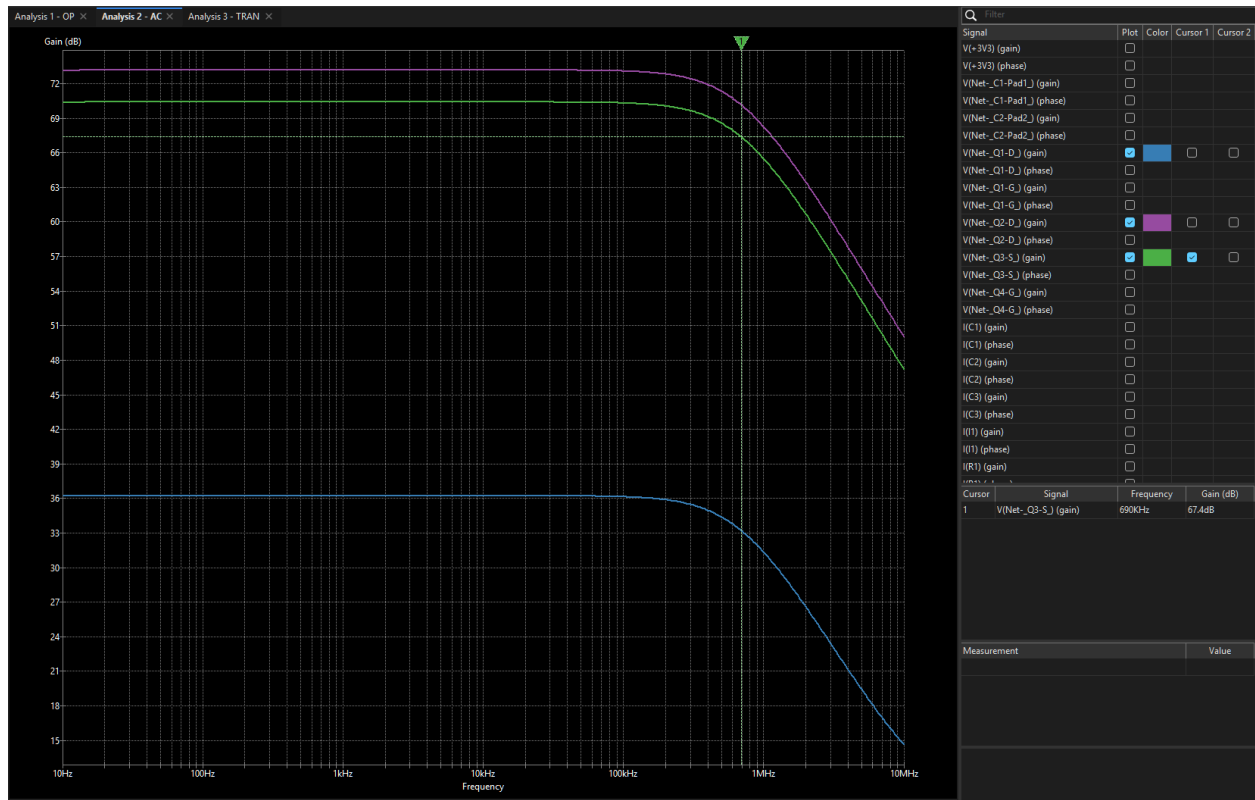


Figure 4: AC analysis for loaded -3dB bandwidth

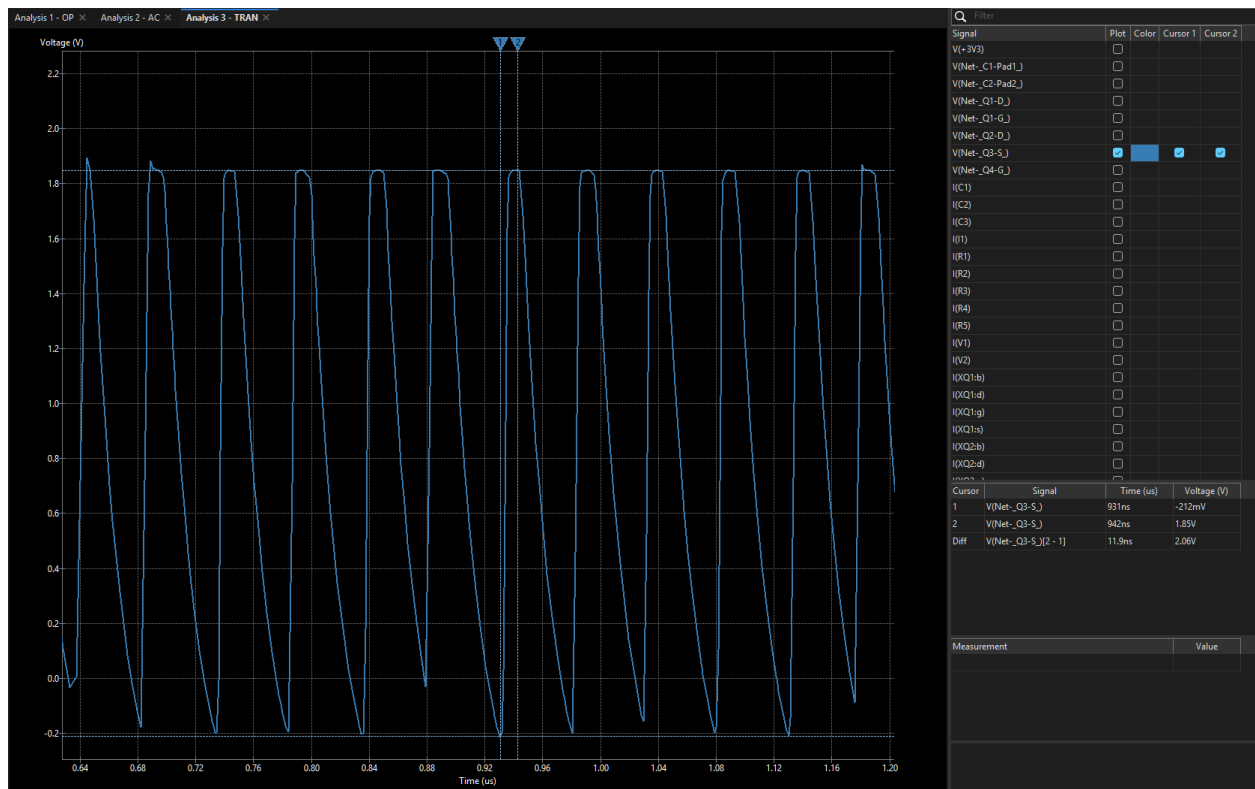


Figure 5: Transient analysis for output swing

5. Analysis & Discussion

The telescopic cascode design had been very effective in achieving the 1 mW power threshold. The overall current draw was minimized by eliminating the need to cascade several common-source stages and using a single high-gain stage. Another important trade-off was observed concerning the output biasing: theoretical analysis showed that the effect of higher PMOS gate voltage would be to move the output node towards higher voltage; in simulations with KiCad, that forced the output node to be pegged at 1.62 V and needed to raise the gate voltage to 1.97 V, to allow the maximum linear output swing. The 18 μm buffer sizing was necessary to counteract the Body Effect to be able to maintain the output resistance at a low value to meet the load sensitivity requirement of less than 10% gain drop.

6. Conclusion

The offered design has been able to achieve all requirements. The circuit achieves a loading gain of 70.4, bandwidth of 690kHz, all consuming only 0.949m using a telescopic cascode with high-value biasing resistors. This architecture shows that the architecture choice is equally important in balancing power and performance as the component sizing.

7. References

ELE 404: “Design Project W2026”. Toronto Metropolitan University.